



DEPARTMENT OF
NETWORKED SYSTEMS
AND SERVICES

Basic operations of quantum informatics

(Műveletek kvantumbitekkel és kvantumregiszterekkel)

Practice

Kvantuminformatikai alkalmazások

Dr. Balázs Solymos Dr. László Bacsárdi

Department of Networked Systems and Services

Faculty of Electrical Engineering and Informatics

bacsardi@hit.bme.hu, solymosb@hit.bme.hu



1th postulate: quantum bit

- Vector in Hilbert space

2nd postulate : logic gates

- Unitary transform
- Elementary logic gates

3rd postulate : Q/C conversion

- Measurement statistics
- Post measurement state

4th postulate : registers

- Tensor product

$$|\varphi\rangle = \sum_{i=0}^{2^n-1} \varphi_i |i\rangle$$

$$U^\dagger \equiv U^{-1}$$

$$P(m \mid |\varphi\rangle) = \langle \varphi | M_m^\dagger M_m | \varphi \rangle$$

$$|\varphi'\rangle = \frac{M_m |\varphi\rangle}{\sqrt{\langle \varphi | M_m^\dagger M_m | \varphi \rangle}}$$

$$|\varphi\rangle = |0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

- Inner product:

$$\langle A|B\rangle \doteq A_1^*B_1 + A_2^*B_2 + \cdots + A_N^*B_N = (A_1^* \ A_2^* \ \cdots \ A_N^*) \begin{pmatrix} B_1 \\ B_2 \\ \vdots \\ B_N \end{pmatrix}$$

- Outer product:

$$|\phi\rangle\langle\psi| \doteq \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_N \end{pmatrix} (\psi_1^* \ \psi_2^* \ \cdots \ \psi_N^*) = \begin{pmatrix} \phi_1\psi_1^* & \phi_1\psi_2^* & \cdots & \phi_1\psi_N^* \\ \phi_2\psi_1^* & \phi_2\psi_2^* & \cdots & \phi_2\psi_N^* \\ \vdots & \vdots & \ddots & \vdots \\ \phi_N\psi_1^* & \phi_N\psi_2^* & \cdots & \phi_N\psi_N^* \end{pmatrix}$$

- Tensor product:

$$|\psi\rangle|\phi\rangle = \begin{bmatrix} \psi_1 \cdot \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} \\ \psi_2 \cdot \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \psi_1\phi_1 \\ \psi_1\phi_2 \\ \psi_2\phi_1 \\ \psi_2\phi_2 \end{bmatrix}$$

- Qubit: $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\langle\psi| = \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix}$, $|\alpha|^2 + |\beta|^2 = 1$
- Inner product: $\langle A|B\rangle \doteq A_1^* B_1 + A_2^* B_2 + \dots + A_N^* B_N = \begin{pmatrix} A_1^* & A_2^* & \dots & A_N^* \end{pmatrix} \begin{pmatrix} B_1 \\ B_2 \\ \vdots \\ B_N \end{pmatrix}$
- Outer product: $|\phi\rangle\langle\psi| \doteq \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_N \end{pmatrix} \begin{pmatrix} \psi_1^* & \psi_2^* & \dots & \psi_N^* \end{pmatrix} = \begin{pmatrix} \phi_1\psi_1^* & \phi_1\psi_2^* & \dots & \phi_1\psi_N^* \\ \phi_2\psi_1^* & \phi_2\psi_2^* & \dots & \phi_2\psi_N^* \\ \vdots & \vdots & \ddots & \vdots \\ \phi_N\psi_1^* & \phi_N\psi_2^* & \dots & \phi_N\psi_N^* \end{pmatrix}$
- Tensor product: $|\psi\rangle\otimes|\phi\rangle = |\psi\rangle|\phi\rangle = \begin{bmatrix} \psi_1 \cdot \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} \\ \psi_2 \cdot \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \psi_1\phi_1 \\ \psi_1\phi_2 \\ \psi_2\phi_1 \\ \psi_2\phi_2 \end{bmatrix}$

Unitary operators and matrices

An invertible complex square matrix U is unitary if its matrix inverse U^{-1} equals its conjugate transpose $U^\dagger$, that is, if $U^\dagger = U^{-1}$.

Properties:

- $UU^\dagger = U^\dagger U = U^{-1}U = I$, where I is the identity.
- Given two complex vectors x and y , multiplication by U preserves their inner product; that is, $\langle x|y \rangle = \langle x|U^\dagger U|y \rangle$.
- $|\det U| = 1$.
- Its eigenspaces are orthogonal.

$$U = \begin{bmatrix} a & b \\ -e^{i\phi}b^* & e^{i\phi}a^* \end{bmatrix}, \quad |a|^2 + |b|^2 = 1$$

$$|0\rangle \frac{|0\rangle + |1\rangle}{\sqrt{2}} =$$

$$(a|0\rangle + b|1\rangle) |0\rangle =$$

$$|0\rangle \frac{|1\rangle + |0\rangle}{\sqrt{2}} =$$

$$|0\rangle (a|0\rangle + b|1\rangle) =$$

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} |0\rangle =$$

$$|1\rangle \frac{|0\rangle + |1\rangle}{\sqrt{2}} =$$

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} |1\rangle =$$

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \frac{|0\rangle + |1\rangle}{\sqrt{2}} =$$

$$|0\rangle \frac{|0\rangle + |1\rangle}{\sqrt{2}} =$$

$$|\psi\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|\phi\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$|\psi\rangle|\phi\rangle = \begin{bmatrix} 1 \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \\ 0 \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} (|00\rangle + |01\rangle)$$

$$|0\rangle \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} (|00\rangle + |01\rangle)$$

$$(a|0\rangle + b|1\rangle) |0\rangle = a|00\rangle + b|10\rangle$$

$$|0\rangle \frac{|1\rangle + |0\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} (|00\rangle + |01\rangle)$$

$$|0\rangle (a|0\rangle + b|1\rangle) = a|00\rangle + b|01\rangle$$

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} |0\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

$$|1\rangle \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} (|10\rangle + |11\rangle)$$

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} |1\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |11\rangle)$$

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

Can the following states form a valid basis?

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Can the following states form a valid basis?

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \quad |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

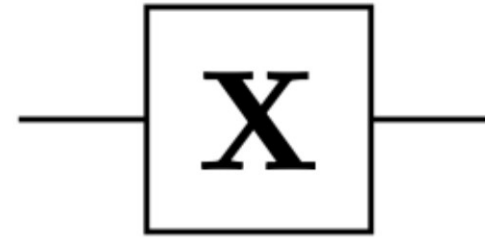
Check orthogonality:

$$\langle + | - \rangle = \frac{1}{\sqrt{2}}\langle 0| - \frac{1}{\sqrt{2}}\langle 1| = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\frac{-1}{\sqrt{2}} = 0$$

Since the result is 0, they are orthogonal, and thus, can form a valid basis.

- Define a qubit, which when measured (in the $|0\rangle, |1\rangle$ basis) will give a result of “1” every time!
- Define a qubit, which gives back 0 with 50% of probability!
- Do the same for measurements in the (in the $|+\rangle, |-\rangle$ basis)

- Pauli X gate, or bit-flip gate
- Symbol:

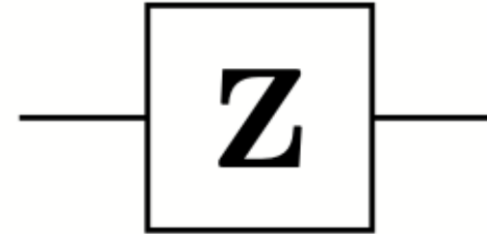


- Matrix: $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

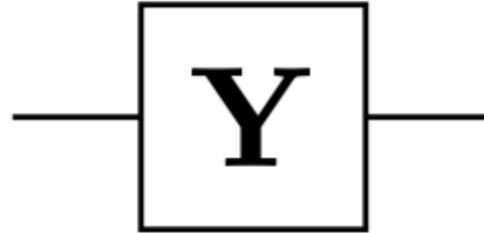
- Its operation on $|\varphi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$

- $X|\varphi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = b|0\rangle + a|1\rangle$

- Pauli Z gate, or phase-flip gate
- Symbol:
- Matrix: $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- Its operation on $|\varphi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$
- $Z|\varphi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = a|0\rangle - b|1\rangle$



- Pauli Y gate
- Symbol:

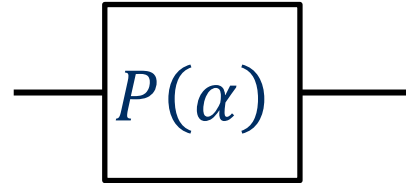


- Matrix: $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

- Its operation on $|\varphi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$

- $Y|\varphi\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = -bi|0\rangle + ai|1\rangle$

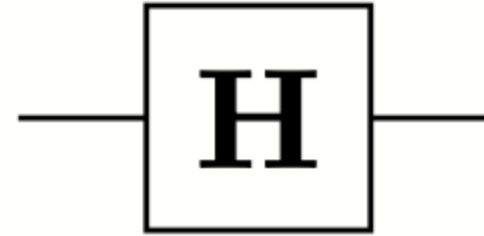
- Phase rotator
- Symbol:



- Alternative symbols (for a particular phase)
- Matrix: $P(\alpha) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{bmatrix}$
- Its operation on $|\varphi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$

- $P|\varphi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = a|0\rangle + e^{i\alpha}|1\rangle$

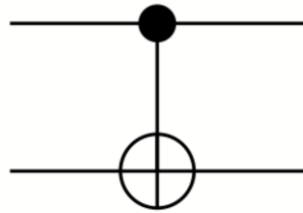
- Hadamard gate
- Symbol:



- Matrix: $H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$
- Its operation on $|\varphi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$

$$\bullet \quad H|\varphi\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = a \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) + b \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} (a + b)|0\rangle + \frac{1}{\sqrt{2}} (a - b)|1\rangle$$

- Symbol:



- Matrix: $CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

- Truth table:

IN		OUT	
x	y	x	$y \oplus x$
0	0	0	$0 \oplus 0 = 0$
0	1	0	$1 \oplus 0 = 1$
1	0	1	$0 \oplus 1 = 1$
1	1	1	$1 \oplus 1 = 0$

$$CNOT (a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle) = a|00\rangle + b|01\rangle + d|10\rangle + c|11\rangle$$

- Take the set of two orthogonal states
(for example: $|\varphi_1\rangle = |0\rangle$, $|\varphi_2\rangle = |1\rangle$)
- How does the projective operator look like?

$$P(m \mid |\varphi\rangle) = \langle \varphi | M_m^\dagger M_m | \varphi \rangle$$
$$|\varphi'\rangle = \frac{M_m |\varphi\rangle}{\sqrt{\langle \varphi | M_m^\dagger M_m | \varphi \rangle}}$$

- Start with two orthogonal states:
 $|\varphi_1\rangle = |0\rangle, |\varphi_2\rangle = |1\rangle$

- Start with two orthogonal states :

$$|\varphi_1\rangle = |0\rangle, |\varphi_2\rangle = |1\rangle$$

- To find M_0 :

$$\langle 0|M_0^\dagger M_0|0\rangle = 1$$

$$\langle 1|M_0^\dagger M_0|1\rangle = 0$$

$$M_0 M_0 = M_0$$

- Searching for M_0 in the following form:

$$\mathbf{M}_0 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

- To find M_0 we have to solve:

$$\langle 0 | M_0^\dagger M_0 | 0 \rangle = 1$$

$$M_0 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\langle 1 | M_0^\dagger M_0 | 1 \rangle = 0$$

$$M_0 M_0 = M_0$$

$$1 = \langle 0 | M_0^\dagger M_0 | 0 \rangle = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^\dagger \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} a^* & c^* \\ b^* & d^* \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow |a|^2 + |c|^2 = 1$$

$$0 = \langle 1 | M_0^\dagger M_0 | 1 \rangle = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^\dagger \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} =$$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} a^* & c^* \\ b^* & d^* \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow |b|^2 + |d|^2 = 0$$

- To find M_0 we have to solve:

$$M_0 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\langle 0 | M_0^\dagger M_0 | 0 \rangle = 1 \longrightarrow |a|^2 + |c|^2 = 1$$

$$\langle 1 | M_0^\dagger M_0 | 1 \rangle = 0 \longrightarrow |b|^2 + |d|^2 = 0$$

$$\left. \begin{array}{l} |b|^2 \geq 0 \\ |d|^2 \geq 0 \end{array} \right\} b = d = 0$$

$$M_0 M_0 = M_0$$

$$\begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} \longrightarrow \begin{array}{l} a^2 = a \\ ac = c \end{array}$$

$$M_0 = \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix}$$

- Solution: $M_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
- Similarly: $M_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

Completeness:

$$\sum_m \mathbf{M}_m^\dagger \mathbf{M}_m = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}$$

Practical notation:

$$M_0 = |0\rangle\langle 0| \text{ and } M_1 = |1\rangle\langle 1|$$

Conclusion

Thus we reached a very simple and practical rule of thumb: *In case we have a set of orthonormal states $\{|\varphi_m\rangle\}$ then the corresponding measurement operators which provide exact differentiation among them can be produced by $M_m = |\varphi_m\rangle\langle\varphi_m|$.*

PLATO'S ALLEGORY OF THE CAVE

